Predicting Next-Day Price Direction of Indian Equities

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Abstract

This report summarizes the methodology, results, and conclusions of a machine learning-based study to predict the next-day price direction of Indian equities using technical indicators. The work involves data preparation, model training (Logistic Regression, Random Forest, XGBoost, SVM, and LightGBM), and evaluation using multiple performance metrics.

1 Workflow and Methodology

Objective: To predict the next-day closing price direction (1 = Up, 0 = Down) using machine learning models trained on technical indicators derived from historical OHLCV data.

Data Preparation

- Dataset: `equities_with_features.csv` containing daily OHLCV data for Indian equities.
- Features engineered:
  - **Trend Indicators:** Simple and Exponential Moving Averages (SMA, EMA)
  - **Momentum Indicators:** Relative Strength Index (RSI), MACD
  - **Volatility Indicators:** Bollinger Bands, Rolling 10-day Volatility
  - **Statistical Features:** Daily Return and Lagged Return

Target Definition

The target variable (`Direction`) is defined as:

$$Direction_t = \begin{cases} 1, & \text{if } Close_{t+1} > Close_t \\ 0, & \text{otherwise} \end{cases}$$

Data Splitting

A time-based split was used: the earliest 80% of data for training and the most recent 20% for testing. This avoids future data leakage and better simulates real-world forecasting.

Evaluation Metrics

Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix, and Rolling Accuracy were used for performance assessment.

Model	Type	Description
Logistic Regression	Linear	Baseline classifier
Random Forest	Ensemble (Bagging)	Captures non-linearity, robust
XGBoost	Ensemble (Boosting)	Strong predictive accuracy
SVM (RBF)	Non-linear Kernel	Handles complex boundaries
LightGBM	Gradient Boosting	Efficient and fast tree-based model

2 Results and Analysis

Model Performance Summary

Model	Accuracy	F1-score	ROC-AUC	Remarks
Logistic Regression	0.55	0.52	0.56	Underfits linear patterns
Random Forest	0.61	0.59	0.63	Stable, interpretable
XGBoost	**0.66**	**0.63**	**0.71**	**Best performer overall**
SVM (RBF)	0.59	0.57	0.61	Moderate, slower training
LightGBM	0.64	0.62	0.69	Comparable to XGBoost, faster

Top Predictive Features

Rank	Feature	Interpretation
1	RSI_14	Measures momentum and overbought/oversold states
2	MACD	Identifies trend strength and reversals
3	Volatility_10	Indicates short-term market uncertainty
4	Lag1_Return	Captures mean reversion or continuation
5	Bollinger Bands	Show volatility-adjusted price extremes

Insights:

- Momentum (RSI, MACD) and volatility features are the most influential.
- XGBoost achieved the best accuracy and generalization.
- Random Forest was highly interpretable with stable results.
- LightGBM matched XGBoost in accuracy but trained faster.
- Linear and SVM models underperformed on non-linear relationships.

3 Conclusions

Summary:

- **Most Predictive Features:** RSI, MACD, and Volatility_10.
- **Best Model:** XGBoost (Accuracy $\approx$ 66%, ROC-AUC $\approx$ 0.71).
- **Key Finding:** Momentum and volatility indicators carry short-term predictive information in Indian equities.

Limitations:

- Excludes macroeconomic and sentiment variables.
- Daily price data is noisy; long-term prediction remains challenging.
- No transaction cost modeling; raw accuracy may not translate to profitability.
- Market non-stationarity requires retraining for sustained performance.

Future Improvements:

1. Implement walk-forward retraining and regime adaptation.
2. Incorporate macroeconomic, volume-based, and sentiment features.
3. Conduct financial backtesting to evaluate real-world performance.
4. Use SHAP values for deeper model explainability.

Final Remark: Momentum and volatility features provide moderate predictability for daily stock movement. Ensemble models like XGBoost and LightGBM outperform simpler baselines, offering a balance of accuracy, robustness, and interpretability.